

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 6: Error Detection & Correction

Course: Computer Networks (Credit Hours: 3)

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1. WHAT IS AN ERROR IN NETWORKS?

When bits travel over transmission cables or wireless signals, interference or noise can flip a **0 into a 1**, or a **1 into a 0**.

Single-Bit Error

Only **1 bit** in the entire data unit gets flipped.

Example Sent: 00000000 → Received: 00000010.

Burst Error

Two or more consecutive bits get flipped due to sudden impulse noise or sparks.

Example Sent: 01000100 → Received: 00111100.

2. THREE EASY ERROR DETECTION TECHNIQUES

How does the receiver know that data got corrupted during transmission?

Technique	How It Works (In Simple Words)	Real-World Use
1. Parity Check (VRC)	Adds 1 extra bit (Parity bit) to make the total count of 1s Even or Odd . If the receiver counts an odd number in an even scheme, an error is caught!	Simple serial communication and keyboard data.
2. Checksum	The sender divides data into equal segments, sums them up, and takes the 1's complement. The receiver adds everything together; if the total is all 1s, data is clean!	Internet Layer & Transport Layer (IP, TCP, UDP headers).
3. Cyclic Redundancy Check (CRC)	Uses binary polynomial division . The sender divides the data bits by a fixed divisor and appends the remainder (CRC). If the receiver divides by the same divisor and gets remainder 0 , there is no error.	Data Link Layer (Ethernet frames, Wi-Fi 802.11).

3. PARITY CHECK: WORKED EXAMPLE

Rule: Even Parity (The total number of 1s in the data block must always be an even number: 2, 4, 6...):

- Data to send: 1 0 1 1 0 0 1 (Count of 1s = 4, already even). Parity bit added = **0** → Transmitted: 1 0 1 1 0 0 1 0.
- Data to send: 1 0 0 1 0 1 0 (Count of 1s = 3, odd). Parity bit added = **1** → Transmitted: 1 0 0 1 0 1 0 1.

Limitation: If **two bits flip simultaneously**, parity check fails to detect the error. That is why LAN networks use **CRC** instead.

4. ERROR DETECTION VS. ERROR CORRECTION

Feature	Error Detection	Error Correction
Action	Only checks if an error occurred ("Yes" or "No").	Finds the exact broken bit and fixes it back (flips it).
Handling	Receiver discards the packet and asks sender to retransmit (ARQ).	Receiver repairs the bit locally using Hamming Code .
Overhead	Low overhead (adds few check bits).	High overhead (requires many redundant check bits).
Best Used For	Reliable wired networks (Ethernet, Fiber).	Satellite links or deep-space where retransmission takes too long.

5. QUICK STUDENT REVIEW QUESTIONS

Q1: Why is CRC preferred over simple Parity in Ethernet LANs?

Ans: Because Parity fails if 2 bits flip together. CRC uses mathematical binary division and can catch almost 100% of burst errors.

Q2: If Error Correction is better, why don't networks use it everywhere?

Ans: Because it requires sending too many extra redundant bits. It is faster and cheaper on fast networks to simply re-send a dropped frame.